



TECH HIGHLIGHTS

LTE Radio Technology Support

- 3GPP Release 15, future proof to support 3GPP up to 17 through SW upgrade
 - CAT-M1: Up to 588 Kbps in DL, and 1119 Kbps in uplink (upto 1.2mbps with Release 17)
 - CAT-NB2: Up to 127 Kbps in DL, and 158 Kbps in uplink (upto 254kbps with Release 17)

Highly Integrated SoC

- Sub 1GHz and 2.4GHz for short range communication and GW
- Integrated PMU with Voltage Regulators and Real-Time clock (RTC)
- ARM cortex-M4 integrated MCU with 1MB NVRAM and 752KB RAM
- Low power always-on sensing hub based ARM Cortex-M0+ integrated MCU
- 2nd Generation Carrier grade iSIM, targeting ELAS+ certification
- Enhanced application layer security- HW based Security Element
- AI low power acceleration for on the edge processing

Application Network Protocols

- Supporting IPv4/IPv6 with TCP/UDP, PPP, FTP, HTTP, TLS, HTTPS, SSL, DTSL, MQTT, CoAP, LWM2M

Operating Temperature

- Maximum temperature: 85c
- Minimum temperature: -40c

Location Services

- Intermittent LTE and GNSS (GPS/GLONASS) navigation for low-cost applications
- Concurrent LTE and L1/L5 GNSS for demanding tracking applications
- Cellular network-based location
- Integrated Wi-Fi SSID based location

LTE Frequency Band Support

- HFDD (Half Duplex FDD) and TDD
- OneSKU™ frequency range:
 - Low Band: 617-960 MHz
 - Middle Band: 1700-2200 MHz
- Complete Supported Band List:
 - LTE-M: 1, 2, 3, 4, 5, 8, 12, 13, 14, 18, 19, 20, 25, 26, 27, 28, 66, 71
 - NB-IoT: 1, 2, 3, 4, 5, 8, 12, 13, 14, 17, 18, 19, 20, 25, 26, 28, 65, 66, 70, 71, 85
 - NTN: n255, n256
- 410-466MHz (bands 31, 72, 73, 87, 88) can be supported using additional RF FE components.

Interfaces

- UART, SPI master and slave, I2C master and slave, PCM/I2S audio, PWM, LED, GPIOs, USIM or eUICC, auxiliary ADC, capture/compare timer, QUAD SPI flash and PSRAM extension ports.

Q Is the chipset capable of satellite connectivity? Please elaborate.

A Yes, our chipsets are capable of satellite connectivity. We work with companies like Skylo and Ligado Networks to develop satellite IT solutions. Currently, this is done with the existing ALT1250 and based on existing modules, but these communications will be supported on the ALT1350 in future as well.

The goal in developing this type of connectivity is to implement satellite connectivity without adding additional hardware. In our vision, we want devices

to be deployed without any coverage holes, so in places where there isn't cellular connectivity or the network is patchy, the devices can communicate over satellite as a backup.

Q Can you comment on the landscape of IoT devices in 2023?

A This is a great question and also pretty easy to answer since the development cycle is about 12-18 months, so we've already seen the designs of everything that is going to be released and shipped this year. We are going to

see an additional shift toward cellular connectivity in the utility segment. Gas meters, water meters, and electricity meters, as well as electric vehicle chargers, are all moving toward more cellular connectivity.

In the asset tracking market, we are seeing more demand for tracking at the pallet and package level, providing more visibility into the supply chain. We are also going to see more regulations coming into play in this area, especially for things like pharmaceutical shipments. **[d]**